ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE H · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice h.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the method and the answer. Accept other correct answers and wording. Symbols, diagrams and circuits may be drawn in different but correct ways.

QUESTION 1

1.1 An inductive sensor uses a magnetic field and detects only metals ✓ – e.g. steel, aluminium ✓. A capacitive sensor uses an electric field and detects almost any material ✓ – e.g. plastic, wood, liquids, paper ✓. (4)

1.2 Any TWO, ✓✓ each: through-beam – the emitter and receiver face each other and an object breaks the beam; retro-reflective – light is sent to a reflector and back, an object breaks the beam; diffuse – light bounces off the object itself back to the receiver. (4)

1.3 Input block ✓ with an example (program buttons, water-level sensor, door switch) ✓; process block ✓ with an example (control board / microcontroller deciding the cycle) ✓; output block ✓ with an example (motor turning the drum, water valve, heater, pump) ✓ (6)

1.4 They tell the controller what is happening in the system (position, temperature, presence) ✓ so that it can make decisions and control the outputs correctly and safely ✓. (2)

1.5 Any TWO, ✓✓ each: microphone – sound to electrical; loudspeaker – electrical to sound; thermocouple – heat to electrical; LDR / solar cell – light to electrical; motor – electrical to mechanical; solenoid – electrical to linear movement. (4)

[20]

QUESTION 2

2.1 Programmable Logic Controller ✓; an industrial computer that reads inputs, runs a stored program and switches outputs to control machines ✓✓. (3)

2.2 Any TWO inputs ✓✓: push buttons, limit switches, proximity sensors, thermostats. Any TWO outputs ✓✓: motors / contactors, solenoid valves, lamps, buzzers. (4)

2.3 I0.0 and I0.1 contacts in series ✓✓ driving coil Q0.0 ✓ (3)

```
|—[ I0.0 ]—[ I0.1 ]———( Q0.0 )—|
```

2.4 I0.2 and I0.3 contacts in parallel ✓✓ driving coil Q0.1 ✓ (3)

```
|———┬———[ I0.2 ]———┬———( Q0.1 )—|
|   └———[ I0.3 ]———┘             |
```

2.5 Any TWO, ✓✓ each: the program can be changed without rewiring; much less wiring; built-in timers and counters; easy fault-finding with status lights; more reliable; smaller. (4)

2.6 $25 = 16 + 8 + 1$ ✓✓ $= 11001_2$ ✓ (3)

[20]

QUESTION 3

3.1 Two lines (A above, B below) with retracted (0) and extended (1) levels ✓✓; steps 1–5 marked ✓; A rises in step 1–2 ✓; B rises in step 2–3 ✓; A falls in step 3–4 ✓; B falls in step 4–5 ✓; neat, labelled ✓ (8)

3.2 "+" means the cylinder extends (goes out) ✓; "−" means it retracts (goes in) ✓. (2)

3.3 It delays a pneumatic signal for a set time before it passes on (using a small reservoir and an adjustable restrictor) ✓✓. Use ✓✓: holding a clamp or press for a few seconds; a delay before a cylinder retracts; glue setting time. (4)

3.4 Any THREE ✓✓✓: manually (push button, lever, foot pedal); mechanically (roller, plunger); pneumatically (pilot air); electrically (solenoid). (3)

3.5 They detect when a cylinder has reached its end position ✓ and send a signal to start the next step ✓, so the sequence runs automatically and in the correct order ✓. (3)

[20]

QUESTION 4

4.1 Any FOUR ✓✓✓✓: DC motor; stepper motor; servo motor; AC induction motor; universal motor; brushless DC motor. (4)

4.2 Swap (reverse) the polarity of the supply to the motor's terminals ✓✓. (2)

4.3 Supply lines ✓; K1 contacts connect + to motor terminal 1 and − to terminal 2 ✓✓; K2 contacts connect + to terminal 2 and − to terminal 1 ✓✓; motor symbol and labels ✓ (6)

4.4 A stepper motor turns in fixed steps for each pulse, usually without feedback (open loop) ✓✓. A servo motor uses feedback (potentiometer / encoder) to move to and hold an exact position (closed loop) ✓✓. (4)

4.5 Any FOUR ✓✓✓✓: voltage; current; power (kW); speed (rpm); frequency; manufacturer; serial number; insulation class / IP rating; connection (star or delta). (4)

[20]

QUESTION 5

5.1 Any FOUR ✓✓✓✓: prevents trips, slips and falls; tools and parts are easy to find; reduces fire risk; keeps dirt out of machines; creates a professional, efficient workplace; emergency exits stay clear. (4)

5.2 Any FOUR tools with uses, ½ + ½ each ✓✓✓✓: screwdrivers – screws on terminals and covers; spanners – nuts and fittings; Allen keys – socket screws; side cutters – cutting cable ties and wires; wire strippers – removing insulation; multimeter – testing circuits; pipe cutter – cutting pneumatic tubing. (4)

5.3 Any SIX in order ✓✓✓✓✓✓: switch off the machine and isolate the electrical supply (lock out and tag); close the air supply and release the pressure; mark / photograph the tube and wire connections; disconnect the coil wiring and the tubes; remove and replace the valve with the correct type; reconnect the tubes and wiring; restore air and power and test the valve and machine; remove the lock and record the work. (6)

5.4 A planned stoppage is scheduled in advance, for example for preventive maintenance or a change of product ✓✓; an unplanned stoppage happens unexpectedly because of a breakdown or fault ✓✓. (4)

5.5 It warns everyone that the equipment is being worked on and must not be switched on ✓, and shows who placed it and why ✓. (2)

[20]

TOTAL: 100

END OF MARKING GUIDELINE